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DIVIDE-AND-CONQUER ALGORITHMS



Divide-and-conquer algorithms are those that split large problems into smaller subproblems, then divide those subproblems into ones that are smaller yet, until they become trivial to conquer. This approach makes recursion an ideal technique to use: the recursive case divides the problem into self-similar subproblems, and the base case occurs when the subproblem has been reduced to a trivial size. One benefit of this approach is that these problems can be worked on in parallel, allowing multiple central processing unit (CPU) cores or computers to work on them.

In this chapter, we'll look at some common algorithms that use recursion to divide and conquer, such as binary search, quicksort, and merge sort. We'll also reexamine summing an array of integers, this time with a divide-and-conquer approach. Finally, we'll take a look at the more esoteric Karatsuba multiplication algorithm, developed in 1960, that laid the basis for computer hardware's fast integer multiplication.